Avionics Flight Display Management (FDM) SRS

Standard: DO-178C Level B Compliant Draft

1. Scope and System Overview

This document defines the primary flight display and avionics sensor telemetry requirements.

2. Functional Requirements

REQ-FDM-001: The flight display unit shall refresh the barometric altitude reading at a frequency of at least 20 Hz with a maximum end-to-end latency of 50 ms.

REQ-FDM-002: The system should update the ground speed display quickly whenever wind gust changes occur.

REQ-FDM-003: The autopilot mode indicator shall display current navigation state, and the unit shall transmit GPS coordinates to the ground control station and emit an audio alarm if link latency exceeds 5

The primary flight display shall present pitch and roll attitude within +/- 0.5 degrees accuracy.

REQ-FDM-004: If the primary Pitot-static airspeed sensor reports an electrical fault or timeout exceeding 100 ms, the software shall transition into secondary synthetic airspeed estimation mode and flag

3. Interface Requirements

REQ-INT-001: The software shall communicate with the Air Data Computer across the ARINC 429 dual-redundant bus at a baud rate of 100 kbps.

REQ-INT-002: The pilot UI shall be modern, intuitive, robust, and easy to use during high-turbulence maneuvers.